

Study on keyhole coupling and melt flow dynamic behaviors simulation of 2219 aluminum alloy T-joint during the dual laser beam bilateral synchronous welding

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ABSTRACT

In this research, the keyhole coupling and melt flow behaviors of 2219 aluminum alloy T-joint during dual laser beam bilateral synchronous welding (DLBSW) process are investigated by simulation, which involves fluid heat transfer and fluid flow. The calculated mesh model of the T-joint is established by simplifying the equal boss of the skin. The combined Gaussian laser heat source is adopted to simulate the laser energy loaded on the base metal during the welding process. The obtained simulation results are reliably matched with the experimental results. Subsequently, not only the coupling process of two keyholes and the evolution characteristics of the coupled keyhole are analyzed, but also the characteristics of molten pool and the flow behaviors of the liquid metal are studied in detail. The results demonstrate that the keyhole profile's oscillations continuously exist, whether before or after the keyhole coupling. The flow direction of the liquid metal near the coupled keyhole wall includes the flow parallel to the keyhole wall and that around the keyhole wall. Besides, the maximum depth of the molten pool slightly lags behind the rear wall of the keyhole. On the coupled keyhole wall and molten pool surface, the melt flow velocities are significantly greater than that inside the molten pool. The vortexes are produced near the upper and lower regions of the coupled keyhole wall.

1. Introduction

The preference of 2219 aluminum alloy for aerospace applications is owing to its excellent performances such as high specific strength, great stress corrosion resistance, reliable performance at low-temperature, good formability and weldability [1,2]. Traditionally, the skin-stringer reinforcement structure, which is manufactured by 2219 aluminum alloy for the application in the spacecraft components, has been produced by the riveting technology [3]. With the development of intelligent and automated manufacturing technology, the riveting technology is gradually replaced by the laser welding technology to manufacture the reinforcement structure by virtue of its excellent weight reduction, high welding quality and efficiency [4,5]. Driven by these superiorities, the dual laser beam bilateral synchronous welding (DLBSW) technology has been firstly applied to manufacture the skin-stringer reinforcement component by Airbus. The welded T-structure has a weight reduction of 18 % as well as a cost reduction of 25 % in contrast to the riveted structure [6,7].

However, numerous works indicate that the movement behaviors of the keyhole and molten pool during the laser welding process are extremely complex so that it has a significant impact on the quality and defects of the welded joint. In order to study the dynamic behaviors of the keyhole and molten pool during the laser welding process, experimental achievements are performed to understand the process directly. Matsunawa et al. [8] studied the dynamics of the keyhole and molten pool during the high-power laser welding of 304 stainless steel by the high speed imaging with X-ray transmission method. Lee et al. [9] adopted the ray tracing method to simulate the laser heat effect on the weld metal for the exploration of the keyhole's formation mechanism during the stationary laser welding process. The results indicated that the keyhole was produced by the movement of the liquid metal induced by vapor recoil pressure and was unstable even in the stationary condition due to its oscillating movement. Zhang et al. [10] observed side profile of the keyhole during the laser deep welding by presenting a butt joint of GG17 glass and stainless steel. Meng et al. [11] explored the effect of T-joint's gap on the plasma plume, keyhole, and molten pool

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dynamics in CO₂ laser welding by high-speed camera. The results shown that the instability of the laser welding process was caused by the energy fluctuations of laser beam in the keyhole's wall.

With the improvement of computational technology, numerical contributions are put forward to describe the mass transfer, heat effect and flow behavior during the laser deep welding process [12,13]. Tan et al. [14] established a 3D transient simulation model to explore the dynamics of the vapor plume, keyhole and molten pool based on the sharp interface method. Sohail et al. [15] researched the influence of gravity on the penetration depth under four spatial positions during fiber laser welding process. It was obtained that gravity slightly affected weld shape and flow structure while it significantly affected the pore characteristics. Ai et al. [16] analyzed the molten pool and keyhole profile during high-power laser welding based on the computational fluid dynamics (CFD) simulation modeling. The model revealed the weld formation process with the consideration of the phase transformation between solid, liquid and vapor.

Moreover, meaningful studies are implemented to explore the liquid metal flow and keyhole behaviors during the laser welding process of aluminum alloy. Pang et al. [17] established a reliable three-dimensional model to simulate the keyhole and molten pool dynamics in the deep penetration laser welding based on successfully experimental verified. Li et al. [18] studied the keyhole evolution and liquid metal behaviors in laser welding of aluminum alloy with different sub-atmospheric pressures. Numerical results demonstrated that compared with that of atmospheric pressure, the keyhole was wider and deeper under sub-atmospheric pressures. Zhang et al. [19] studied the characteristics of molten pool behaviors, keyhole evolution and root hump formation during laser penetration welding of aluminum alloy by modeling with volume of fluid (VOF) and beam-tracing methods. The achievements indicated that the root hump was formed by overheated sagging with low surface tension stagnating at the bottom position.

According to the above undeniable investigations, the keyhole and formation mechanism, molten pool and flow behavior, plasma action, pore induced by the keyhole during the laser welding process are studied comprehensively with numerical and experimental methods. However, there is few research about the keyhole and liquid metal flow during DLBSW of the aerospace aluminum alloy T-joint. The main studies focus on microstructure, welding defects, mechanical properties, residual stress and deformation after DLBSW [20–24]. Furthermore, the behaviors of the fluid flow and the keyhole during the DLBSW process still has a significant effect on the welding quality.

In order to investigate keyhole coupling and liquid melt flow behaviors of 2219 aluminum alloy T-joint during the DLBSW process, a three-dimensional multiphase transient model is established to simulate T-joint's DLBSW process in the present investigation. The simulation results are reliably matched with the experimental results. By performing the evolution of two keyholes and the melt flow field of molten pool, the characteristics of combined molten pool with coupled keyhole are investigated detailly. It has important basic significance to further understand the DLBSW process of the aluminum alloy T-joint used in the aerospace service structure.

2. Experimental details

The chemical composition of the 2219 aluminum alloy (base metal) used in this research is given in Table 1. The grade is developed by the American Aluminum Association (AA) standard.

Rolled panel of 3.5 mm thick and L-shaped extruded section of 4 mm thick machined by the base metal are applied as the skin and stringer,

Table 1
Chemical composition of 2219 aluminum alloy (wt.%).

Cu	Mn	Fe	Ti	V	Zn	Si	Zr	Al
6.48	0.32	0.23	0.06	0.08	0.04	0.49	0.2	Balance

respectively. In addition, two equal bosses ($1.5 \times 1.5 \text{ mm}^2$) are pre-fabricated in the weld location of the skin to play a role in replacing filled wire and stabilizing the stringer during the DLBSW process. Fig. 1 depicts schematic diagram of the T-joint's DLBSW with dimension details. The formation of weld seam is occurred at the contact area of the stringer and skin under the energy of the two laser beams on the bosses.

It is exhibited in Fig. 2 that the experimental apparatuses include two laser welding heads assembled on two KUKA robots, a shielding gas system, a welding platform, an Acuteye high-speed camera system and a laser device. The laser device is the TruDisk 12003-disk solid laser, which could adjust the laser power between 0.24 kW and 12 kW.

3. Numerical simulation modeling

3.1. Modeling assumptions

It is complicated to involve the heat transfer and liquid metal flow with phase transformation as well as keyhole evolution in the DLBSW process. Hence, some assumptions are required to reliably simplify the numerical model for the simulation in this paper. The modeling assumptions are expressed as follows:

- (1) The liquid metal in molten pool is an incompressible Newtonian fluid with a state of laminar flow.
- (2) The influence of shielding gas on the liquid metal flow behavior is ignored during the DLBSW process.
- (3) The base metal is assumed isotropic homogeneous. The material properties parameters are constant, but solid and liquid parameters are different.
- (4) The mesh at the bosses is simplified according to the cross-sectional shape of the T-joint after welding.
- (5) The welding process is transient, and the initial temperature of the T-joint is set to 300 K.

3.2. Governing equations

The governing equations of fluid flow dynamic during the DLBSW process are divided into mass, momentum, and energy conservation in Cartesian coordinate system. They are drawn by following descriptions [25]:

Continuity equation:

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = 0 \quad (1)$$

Momentum equation:

X Direction :

$$\begin{aligned} & \frac{\partial(\rho u)}{\partial t} + \frac{\partial(\rho uu)}{\partial x} + \frac{\partial(\rho uv)}{\partial y} + \frac{\partial(\rho uw)}{\partial z} \\ &= -\frac{\partial P}{\partial x} + \frac{\partial}{\partial x} \left(\mu \frac{\partial u}{\partial x} \right) + \frac{\partial}{\partial y} \left(\mu \frac{\partial u}{\partial y} \right) + \frac{\partial}{\partial z} \left(\mu \frac{\partial u}{\partial z} \right) + S_u \end{aligned} \quad (2)$$

Y Direction :

$$\begin{aligned} & \frac{\partial(\rho v)}{\partial t} + \frac{\partial(\rho vu)}{\partial x} + \frac{\partial(\rho vv)}{\partial y} + \frac{\partial(\rho vw)}{\partial z} \\ &= -\frac{\partial P}{\partial y} + \frac{\partial}{\partial x} \left(\mu \frac{\partial v}{\partial x} \right) + \frac{\partial}{\partial y} \left(\mu \frac{\partial v}{\partial y} \right) + \frac{\partial}{\partial z} \left(\mu \frac{\partial v}{\partial z} \right) + S_v \end{aligned} \quad (3)$$

Z Direction :

$$\begin{aligned} & \frac{\partial(\rho w)}{\partial t} + \frac{\partial(\rho wu)}{\partial x} + \frac{\partial(\rho wv)}{\partial y} + \frac{\partial(\rho ww)}{\partial z} \\ &= -\frac{\partial P}{\partial z} + \frac{\partial}{\partial x} \left(\mu \frac{\partial w}{\partial x} \right) + \frac{\partial}{\partial y} \left(\mu \frac{\partial w}{\partial y} \right) + \frac{\partial}{\partial z} \left(\mu \frac{\partial w}{\partial z} \right) + S_w \end{aligned} \quad (4)$$

Energy equation:

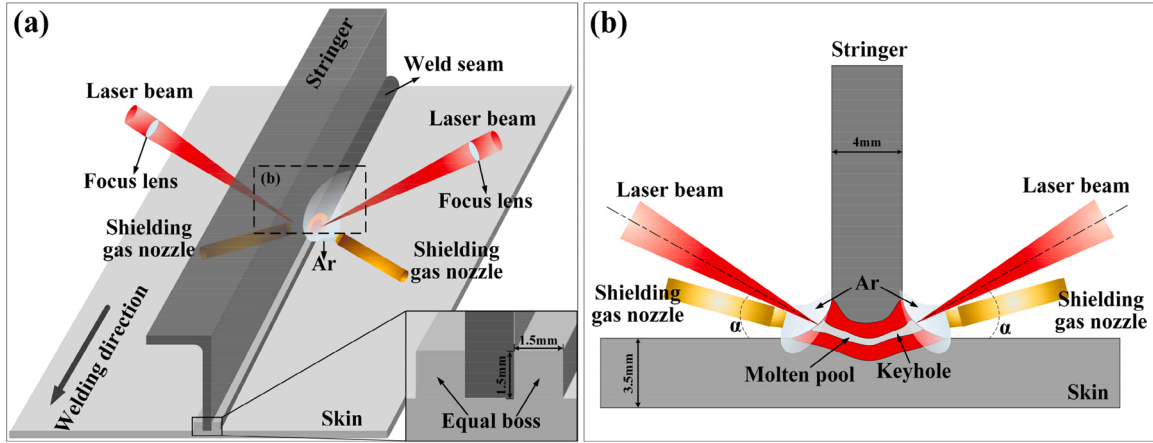


Fig. 1. Schematic diagram of the T-joint's DLBSW with dimension details.

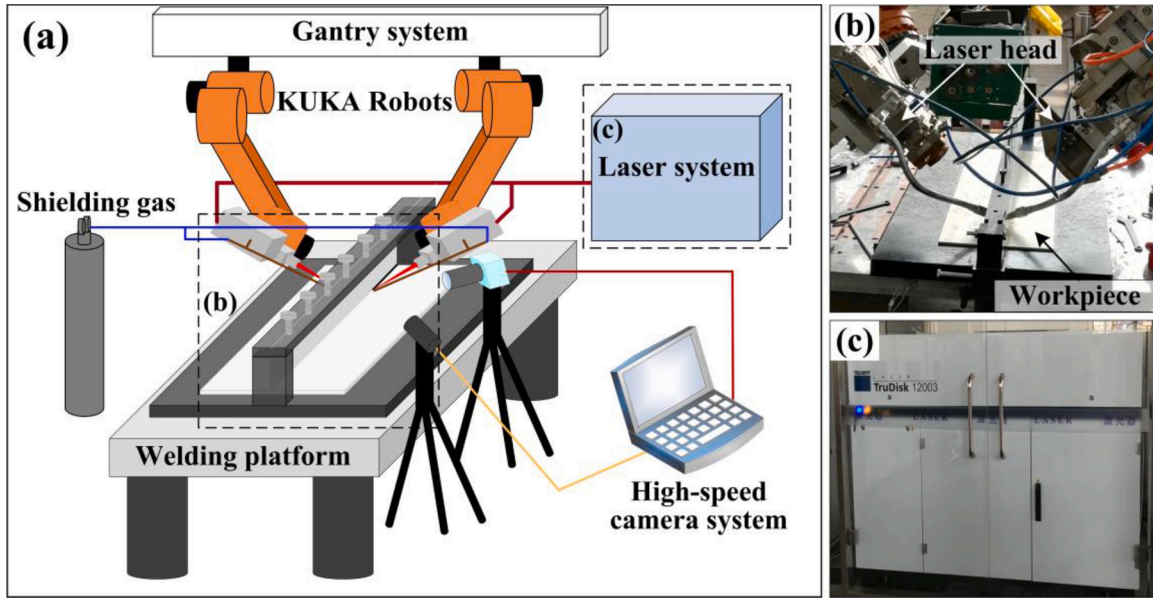


Fig. 2. Schematic diagram and equipment of DLBSW in the experiment.

$$\begin{aligned} & \frac{\partial(\rho H)}{\partial t} + \frac{\partial(\rho u H)}{\partial x} + \frac{\partial(\rho v H)}{\partial y} + \frac{\partial(\rho w H)}{\partial z} \\ &= \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) + S_E \end{aligned} \quad (5)$$

where u , v , and w are respectively the velocity components in the x , y , and z direction. ρ is the density of liquid fluid, and P is the ambient pressure. μ is the viscosity. k is the thermal conductivity of the material. H is the mixing enthalpies. S_u , S_v and S_w are the momentum source terms in the x , y , and z direction, respectively, and S_E is the energy source term.

Moreover, the momentum source terms and energy source term are the key factors to calculate velocity and temperature fields in the molten pool and free surface deformation of the keyhole during the welding process. For the momentum source terms, S_u , S_v and S_w could be written as follows [26,27]:

$$S_u = -C \frac{(1-f_L)^2}{f_L^3 + B} (u - u_0) \quad (6)$$

$$S_v = -C \frac{(1-f_L)^2}{f_L^3 + B} v \quad (7)$$

$$S_w = -C \frac{(1-f_L)^2}{f_L^3 + B} w + \rho g \beta (T - T_l) \quad (8)$$

$$f_L = \begin{cases} 0 & T < T_s \\ \frac{T - T_s}{T_l - T_s} & T_s < T < T_l \\ 1 & T > T_l \end{cases} \quad (9)$$

where the first term on the right side of the above each formula is the momentum loss of the mushy zone calculated according to the Carman-kozeny formula [28]. f_L is the volume fraction of the liquid phase. T_l and T_s respectively represent the liquidus and solidus temperature of the base metal. B is a very small constant introduced to prevent the denominator of the term from being zero. C is the Darcy constant related to the volume fraction of the liquid phase [29]. Besides, the second term is the buoyancy source term. β is the thermal expansion coefficient and T_l is the liquidus temperature.

For the energy source term, S_E could be expressed as below [30]:

$$S_E = Q - m_v L_v - \left[\frac{\partial}{\partial t}(\rho \Delta H) + \frac{\partial}{\partial x}(\rho u \Delta H) + \frac{\partial}{\partial y}(\rho v \Delta H) + \frac{\partial}{\partial z}(\rho w \Delta H) \right] + \frac{\partial}{\partial x}(\rho u_0 \Delta H) \quad (10)$$

where Q is the laser heat source. The second term represents the energy conservation induced by evaporation of the liquid metal. m_v is the mass transfer, and L_v is the latent heat of evaporation. The remaining part is the latent heat source release during melting and solidification process.

3.3. Boundary conditions

The schematic diagram of solution model is displayed in Fig. 3. During solution modeling for the T-joint's DLBSW process, the boundary conditions are essential to keep the calculated accuracy with the involvement of complicated physics phenomena.

The calculated model is divided into gas and metal domains, which are represent by blue line zone and black line zone, respectively. The whole domain at the beginning of the solution is loaded at ambient pressure and room temperature. The top surfaces of gas domain are all determined as the pressure-inlet in this model, and the other surfaces of it are determined as the pressure-outlet. The surfaces of T-joint are performed as wall boundaries with heat convection and heat radiation. The thermal boundary condition is described as follows formula [31]:

$$k \frac{\partial T}{\partial n} = -h_c(T - T_{ref}) - \varepsilon \sigma (T^4 - T_{ref}^4) \quad (11)$$

where T and T_{ref} are respectively the T-joint's temperature and ambient temperature. $\vec{n}$ is the normal vector along the wall boundary. h_c is the convection coefficient. ε is the surface radiation emissivity, and σ is the Stephan-Boltzmann constant.

For the free surface of keyhole, the Fresnel absorption, convection, radiation and metal evaporation are all involved in the thermal boundary conditions, which are drawn as follows [32]:

$$k \frac{\partial T}{\partial n} = q_{abs} - \varepsilon \sigma (T^4 - T_{ref}^4) - h_c(T - T_{ref}) - W_{evp} H_v \quad (12)$$

where q_{abs} is the absorbed laser energy, ε is the thermal emissivity of the base metal. W_{evp} and H_v are the evaporation rate and evaporation latent of the liquid metal, respectively. In addition, the temperatures of the metal surfaces are all determined as the ambient temperature (300 K) at the initial moment.

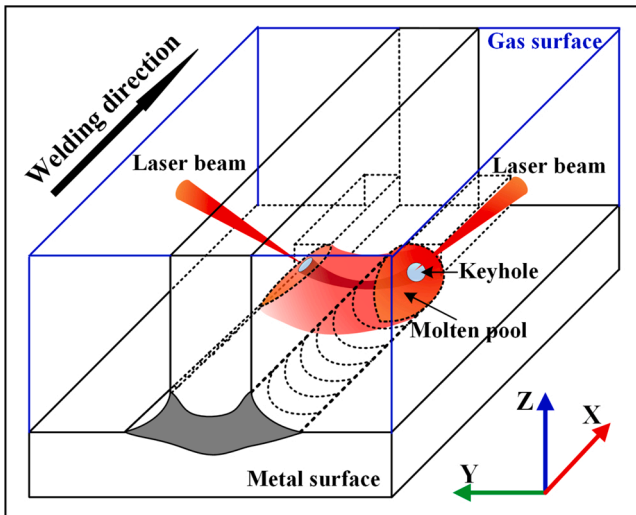


Fig. 3. The solution model and boundary conditions for calculation.

3.4. Laser heat source model

Due to the effect of the laser beam's energy, the base metal is melted and evaporated to form the keyhole, which generates the plasma cloud with extremely high energy above the molten pool [33]. It could be equivalent to a surface heat source above molten pool because of continuous radiate energy on the molten pool surface. According to the absorption mechanism comprised of Fresnel and inverse bremsstrahlung absorption in the keyhole, it is indicated that the inside of the keyhole filled with metallic steam could be described as a body heat source with enormous energy vaporizing the keyhole's wall [34,35]. Therefore, a combined heat source model is employed to simulate the laser beam's energy transfer in the paper. Its schematic diagram is presented in Fig. 4. This combined heat source is consisted of the Gaussian surface heat source and the Rotary-Gaussian body heat source.

The surface heat source model is depicted as follows:

$$q_s(x, y) = \frac{\eta_1 Q_s}{\pi r_s^2} \exp\left(-\frac{\alpha(x^2 + y^2)}{r_s^2}\right) \quad (13)$$

where η_1 is the concentration coefficient of the surface heat source. Q_s is the surface heat source effective energy, and r_s is the heat source effective radius. α is the correction coefficient adjusted by numbers of simulation until the results agreed well with the experimental result.

The body heat source model is written as follows:

$$q(x, y, z) = \frac{9\eta_2 Q_v}{\pi h r_s^2 (1 - e^{-3})} \exp\left(\frac{-9(x^2 + y^2)}{r_s^2 \ln(h/z)}\right) \quad (14)$$

where η_2 is the concentration coefficient of body heat source. Q_v is the effective energy of body heat source. h is the effective depth of body heat source. Besides, the relationship between the two sub-heat source is able to be drawn as follows:

$$Q\eta = Q_s + Q_v \quad (15)$$

where Q is total laser power, η is the absorptivity of base metal.

The parameters of the combined heat source is list in Table 2 based on other published research and laser equipment parameters.

3.5. Coordinate system transformation

According to the spatial characteristics of the DLBSW, two laser beams are symmetrically distributed on both sides of the stringer with a same angle θ to the vertical plane. Therefore, the coordinate system of the combined heat source must be transformed accordingly. In this paper, the intersection line of the horizontal plane where the stringer is in contact with the skin and the mid-plane in the thickness direction of the stringer is set as the X axis. Therefore, the combined heat sources on the left and right sides need to rotate counterclockwise and clockwise by the angle θ , respectively. The calculation equations of the coordinate movement and rotation can be expressed as follows:

$$\begin{cases} x_{rl} = x_l \\ y_{rl} = \cos(\theta)(y - y_l) - \sin(\theta)(z - z_l) \\ z_{rl} = \cos(\theta)(z - z_l) + \sin(\theta)(y - y_l) \end{cases} \quad (16)$$

$$\begin{cases} x_{rr} = x_r \\ y_{rr} = \cos(\theta)(y - y_r) + \sin(\theta)(z - z_r) \\ z_{rr} = \cos(\theta)(z - z_r) - \sin(\theta)(y - y_r) \end{cases} \quad (17)$$

where x, y, z are the coordinates of two combined heat sources before the movement of coordinates. $x_l, y_l, z_l, x_r, y_r, z_r$ are the coordinates of two combined heat sources after the movement of coordinates. $x_{rl}, y_{rl}, z_{rl}, x_{rr}, y_{rr}, z_{rr}$ are the coordinates of two combined heat sources after the rotation of coordinates.

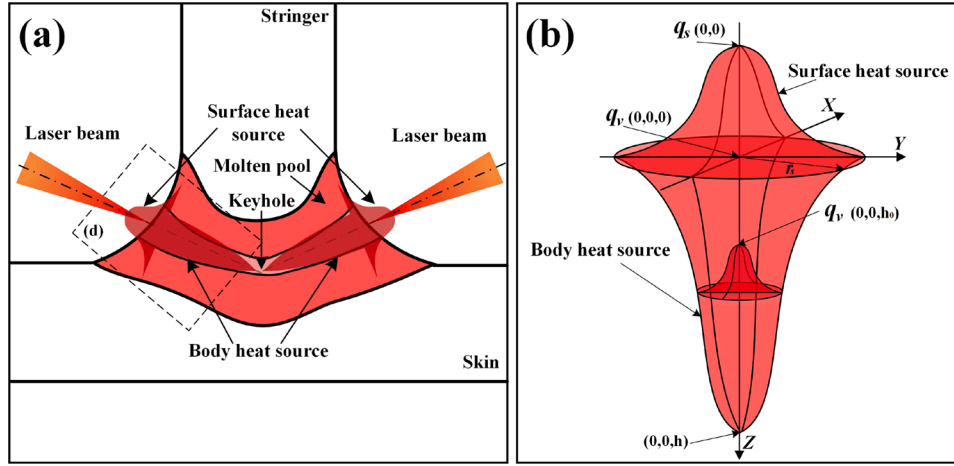


Fig. 4. Schematic diagram of the heat source model used in the simulation.

Table 2

The parameters of the combined heat source for DLBSW of 2219 aluminum alloy T-joint.

Name	Symbol	Unit	Value
Concentration coefficient of the surface heat source	η_1	/	0.2
Surface heat source effective energy	P_s	W	1125
Heat source effective radius	r_s	mm	0.225
Correction coefficient	α	/	1
Concentration coefficient of body heat source	η_2	/	0.8
Effective energy of body heat source	P_v	W	3375
Effective depth of body heat source	h	mm	4.5
Total laser power	P	W	4500
Absorptivity of base metal	η	/	0.86

3.6. Keyhole surface tracking

The flow physics of liquid-gas interface is tracked by VOF method [36]. The equation of the method is drawn as follow equation:

$$\frac{\partial F}{\partial t} + u \frac{\partial F}{\partial x} + v \frac{\partial F}{\partial y} + w \frac{\partial F}{\partial z} = 0 \quad (18)$$

where u , v , w are the flow velocities along x , y , z directions, respectively. The liquid-gas interface in the calculated model could be solved by Eq. (18). When $F = 0$, the domain is occupied by vapor gas. If $F = 1$, the domain is occupied by metal liquid. If $0 < F < 1$, the domain locates on the phase interface which includes both liquid and gas.

3.7. Driving forces

In this simulation, the behaviors of the melt flow and keyhole in the molten pool are affected by the vapor recoil pressure, surface tension, hydrostatic and hydrodynamic pressure. The mainly driving forces involved in the model include the vapor recoil pressure and the surface tension. On the keyhole wall, the equilibrium of pressure is expressed as follows [37]:

$$P = P_\gamma + P_r + P_g + P_h \quad (19)$$

where P is total pressure. P_γ , P_r , P_g , and P_h are the surface tension, vapor recoil pressure, hydrostatic and hydrodynamic pressure, respectively.

P_r is an essential force to form the keyhole during the DLBSW process. It could be described mathematically as follows [38]:

$$P_r = AB_0 T_w^{-1} \exp\left(\frac{-M_a H_v}{N_a k_b T_w}\right) \quad (20)$$

where A is the coefficient related to pressure. B_0 is the evaporation

constant related to the base metal. T_w is the temperature of the keyhole wall. M_a is the atomic mass of the alloy. H_v is the latent heat of evaporation. N_a is Avogadro constant and k_b is the Boltzmann constant.

Besides, the surface tension of base metal is related to the temperature variation. The surface tension is represented by the follow function [39,40]:

$$P_\gamma = K[\delta_0 + A_\delta(T - T_l)] \quad (21)$$

where K represents the free surface curvature related to pressure. δ_0 represents the coefficient of the surface tension. A_δ represents the temperature gradient of the surface tension. T_l represents the liquidus temperature of the base metal.

3.8. Computational domain and material properties

In this study, the length range of 16 mm in the welding direction is taken as the computational domain, which includes gas and metal. The details of the dimensions and mesh are shown in Fig. 5. It is worth noting that the bosses of the metal have been simplified according to the profile of the weld seam after the DLBSW process. The ultimate mesh model is divided into 367770 mesh cells with the minimum cell of $0.15 \times 0.15 \times 0.15 \text{ mm}^3$, which is set to guarantee the computational accuracy between the simulation and the experimental results. The main thermal physical properties of the base metal utilized in the paper are given in Table 3, which is obtained by the professional calculation software and related reference [41,42].

4. Results and discussion

4.1. Simulation model validation

Based on the welding process parameters given in Table 4, the compared results between simulation and experiment of the T-joint's DLBSW are clearly displayed in Fig. 6.

It is observed from Fig. 6 that the simulated weld profile (white dotted line) is in good agreement with that (red dotted line) of the experiment results. The size parameters of the weld seam and molten pool are exhibited in Table 5. It is obtained that the values of half weld width, weld depth, weld metal width and molten pool length in simulation and experiment are relatively similar. It is indicated that the calculated model is reliable in this research.

4.2. Keyhole characteristics

It is exhibited in Fig. 7(a–f) that the profiles of molten pool and keyholes are evolutionary in cross-section view with different moments.

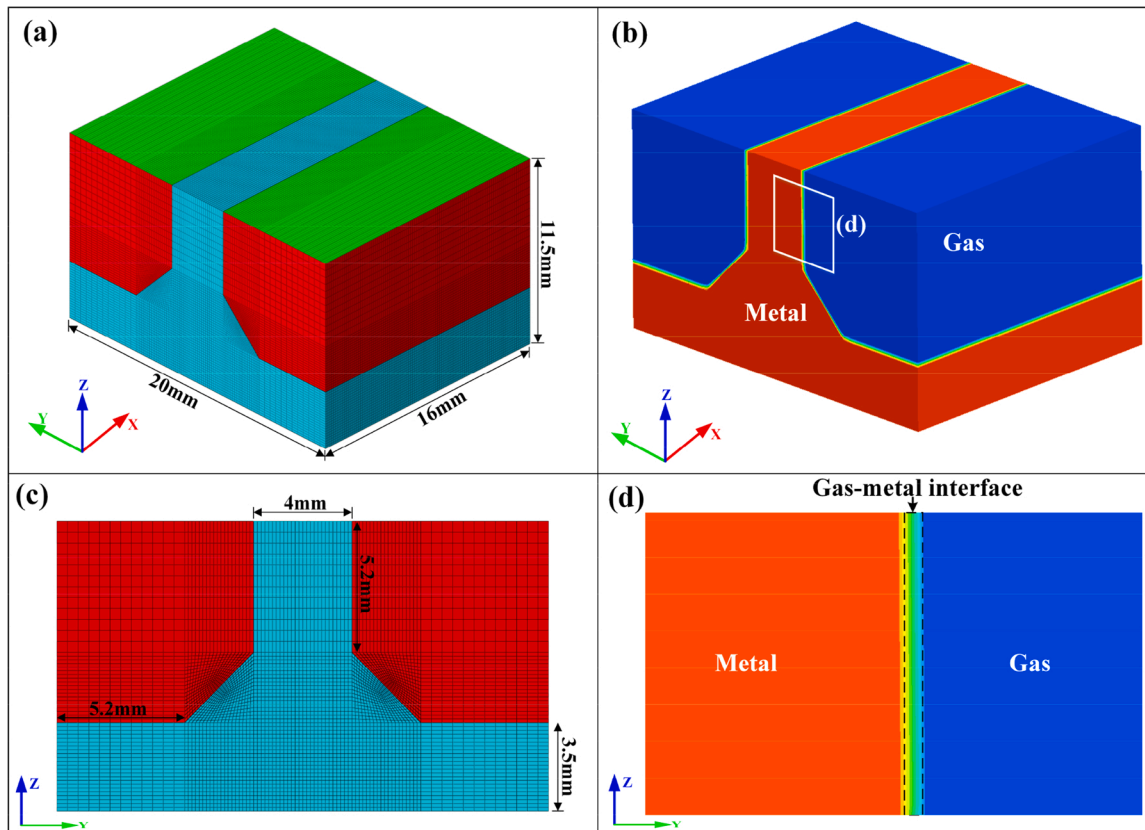


Fig. 5. Mesh model and computational domain of the T-joint. (a) Mesh model of the computational domain. (b) Computational domain with phase distribution. (c) Cross-section of mesh model. (d) Schematic diagram of gas-metal interface marked in (b).

Table 3

Thermal physical properties of 2219 aluminum alloy.

Name	Symbol	Unit	Value
Solidus density	ρ_s	kg m^{-3}	2700
Liquidus density	ρ_l	kg m^{-3}	2400
Thermal conductivity of solid	k_s	$\text{W m}^{-1} \text{K}^{-1}$	238
Thermal conductivity of liquid	k_l	$\text{W m}^{-1} \text{K}^{-1}$	100
Specific heat capacity of solid	C_{ps}	$\text{J kg}^{-1} \text{K}^{-1}$	871
Specific heat capacity of liquid	C_{pl}	$\text{J kg}^{-1} \text{K}^{-1}$	1060
Latent heat of fusion	H_m	J kg^{-1}	3.87×10^5
Latent heat of evaporation	H_v	J kg^{-1}	1.08×10^7
Coefficient of surface tension	δ_0	N m^{-1}	0.914
Temperature gradient of surface tension	A_δ	$\text{N m}^{-1} \text{K}^{-1}$	-0.35×10^{-3}
Dynamic viscosity	μ	$\text{kg m}^{-1} \text{s}^{-1}$	0.006
Coefficient of thermal expansion	β		2.36×10^{-5}
Solidus temperature	T_s	K	820
Liquidus temperature	T_l	K	930
Convective heat transfer coefficient	h_c	$\text{W K}^{-1} \text{m}^{-2}$	15
Thermal emissivity	ε		0.4
Ambient temperature	T_{ref}	K	300

Table 4

DLBSW process parameters (each side) utilized in simulation and experiment of T-joint.

Parameters	Value
Laser power (W)	4500
Welding speed (m/min)	2.5
Laser incidence Angle($^\circ$)	30
defocusing amount(mm)	0
Shielding gas flow(L/min)	15

The liquid metal of the molten pool irradiated by laser beam is

evaporated dramatically to form the keyhole due to the effect of vapor recoil pressure, as displayed in Fig. 7(a). This phenomenon is determined by the energy distribution of the combined heat source. The energy density near the central axis of the heat source is much higher than that at the edge of the heat source. Therefore, the liquid metal in the central area of the molten pool is primarily evaporated, and the generated vapor recoil pressure has a "digging" effect along the depth of the molten pool, thereby forming a keyhole.

The temperature of the keyhole area is significantly higher than that of its surrounding area. Under the effect of heat conduction, the heat of the molten pool comes from the combined effect of surface heat source and keyhole heat transfer. Subsequently, the volumes of molten and evaporated metal gradually increase with the continuous absorption of laser energy by base metal. The distance between the bottom tips of two keyholes decreases until the keyhole walls mutually contact to form a coupled keyhole, as shown in Fig. 7(f).

Aiming to explicitly explore the keyholes' evolution and their coupling behavior, the three-dimensional views of each moment are presented in Fig. 8(a–f), respectively. The keyhole profile begin to oscillate to keep the force balance of the keyhole wall, resulting in humps formed on the keyhole wall, as displayed in Fig. 8(b–d). Part of hump surface could be directly irradiated by incident or reflected laser beam. Due to the effect of Fresnel absorption, the surface temperature rises sharply and the evaporation becomes intensive, which causes the forces on the hump surface to be rebalanced so that lead to the oscillation of keyhole profile. In addition, the profile of combined molten pool is approximately symmetrical with the stringer but that of the keyholes are not completely symmetrical while the space and energy distribution of the laser beams on both sides are symmetrical with the stringer. It is demonstrated that the force balance of the keyhole wall on both sides of the stringer center is irregularly evolved during the DLBSW process.

Fig. 9 describes the schematic diagram of keyhole evolution and

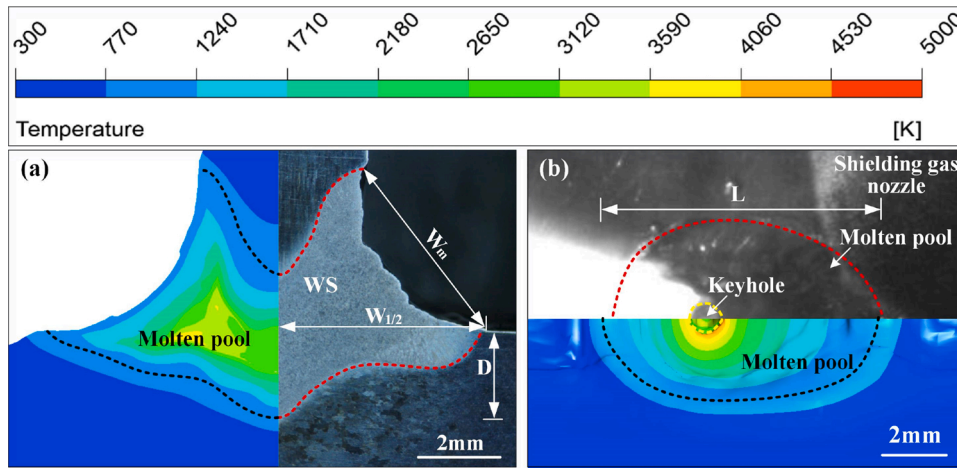


Fig. 6. The profile comparison of simulation results and experimental results.

Table 5

Size comparison of weld seam and molten pool in simulation and experiment.

Parameters	Simulated values(mm)	Experimental values(mm)
Half weld width ($W_{1/2}$)	5.46	5.07
Weld depth (D)	2.11	2.13
Weld seam width (W_m)	5.38	4.94
Molten pool length (L)	8.34	8.14

coupling behavior. The keyhole wall is subjected to the vapor recoil pressure (P_v), surface tension (F_s), hydrostatic pressure (P_m), hydrodynamic pressure (F_v), and vapor-metal friction force (F_f). The force condition of the hump formed by the evolution of the keyhole is shown in Fig. 9(b). Due to the effect of the vapor recoil pressure, the vaporization location of hump generates the metallic vapor jet. The hump subjected to the combined effect of these volume and surface forces gradually moves along keyhole wall and disappears with evaporation, as shown in

Fig. 9(e). On the tip of keyhole, the vapor recoil pressure is also perpendicular to the cut surface at the location, as presented in Fig. 9(c). Due to the continuous evaporation produced by the liquid metal on the keyhole wall, tips of two keyholes on both sides gradually approach and contact under the effect of the combined force. Ultimately, a coupled keyhole is formed with the rebuilding of the force condition at the coupled area of the keyhole exhibited in Fig. 9(f). It is known that the profile of the heat source is fixed, but the force condition of the keyhole wall is unfixed considering the surface tension and vapor recoil pressure will be significantly affected by temperature variation. Therefore, the force condition on the wall of two keyholes or coupled keyhole is not constant during the whole welding process so that the hump and oscillation might be occurred continuously whether before or after the keyhole coupling.

According to the geometric dimensions of molten pool and keyhole exhibited in Fig. 10, the surface area of molten pool is extremely small at the initial stage of the welding process. Its length is approximately 2

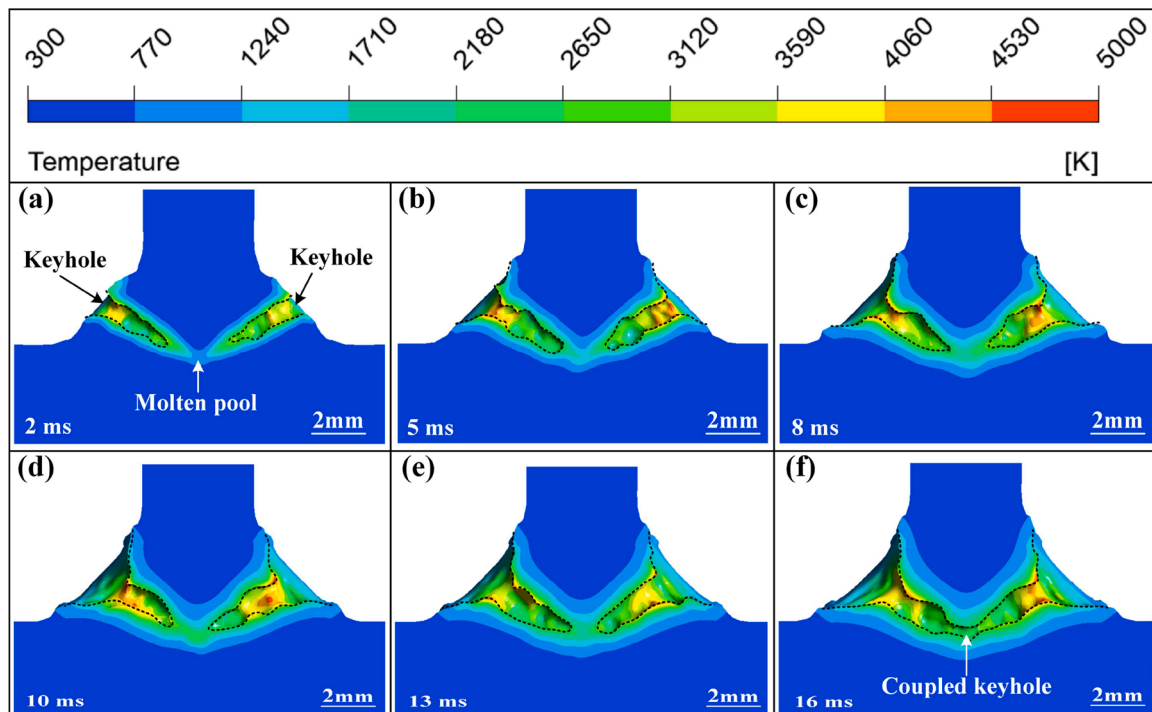


Fig. 7. Cross-sectional profile of molten pool and keyhole until keyhole coupling. (a) 2 ms. (b) 5 ms. (c) 8 ms. (d) 10 ms. (e) 13 ms. (f) 16 ms.

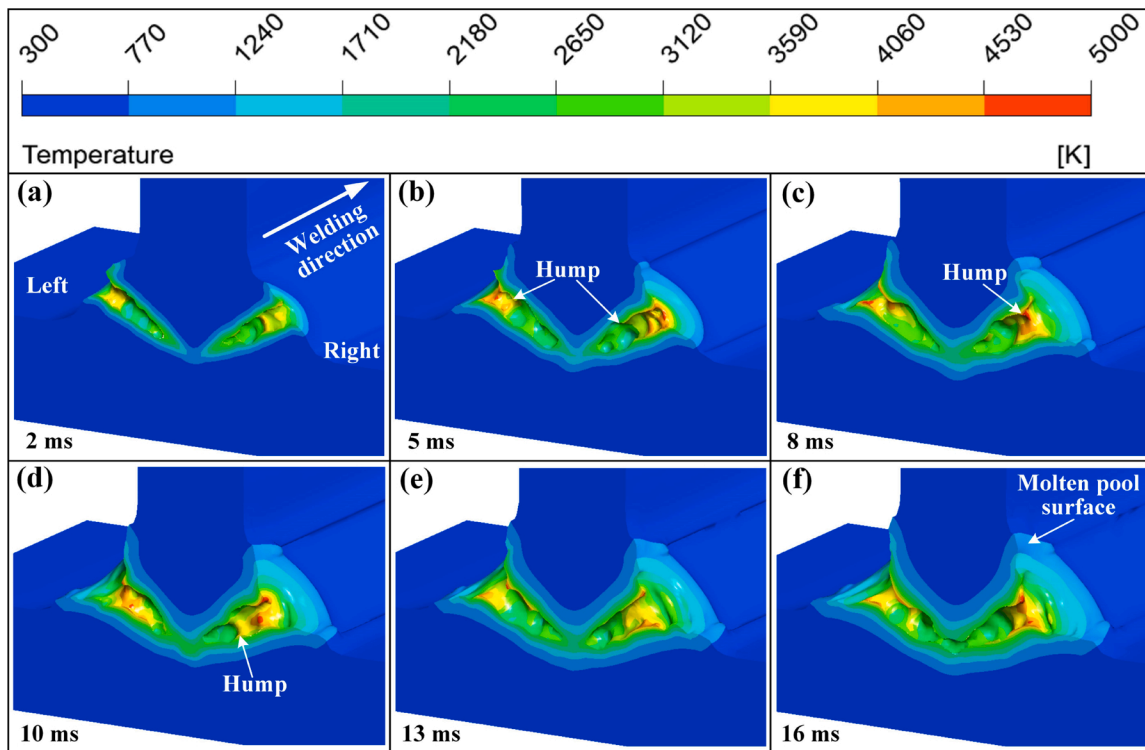


Fig. 8. Three-dimensional profile of molten pool and keyhole until keyhole coupling. (a) 2 ms. (b) 5 ms. (c) 8 ms. (d) 10 ms. (e) 13 ms. (f) 16 ms.

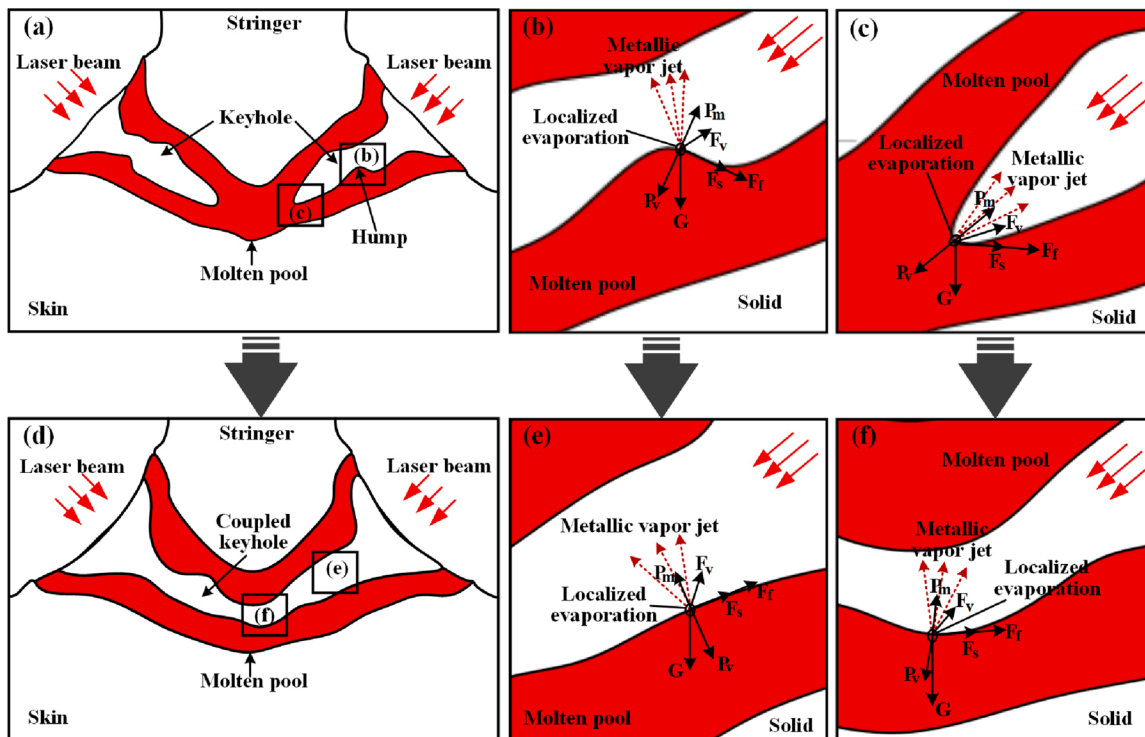


Fig. 9. Schematic diagram of keyhole evolution and coupling process. (a) The keyhole uncoupled stage. (b) The force analysis on the location of hump marked in (a). (c) The force analysis on the tip of keyhole marked in (a). (d) The keyhole coupled stage. (e) The force analysis on the location marked in (d). (f) The force analysis on the location of keyhole coupled marked in (d).

mm, which is about four times of the keyhole opening diameter at the same time. However, the surface area of molten pool expands rapidly due to energy absorption by the base metal. The length of molten pool is close to 5 mm at the moment of 16 ms, which is nearly three times of the

keyhole opening diameter. It is concluded that the molten pool size and the keyhole opening diameter on both sides evolve drastically before the keyhole coupling while size difference of them on both sides is inappreciable.

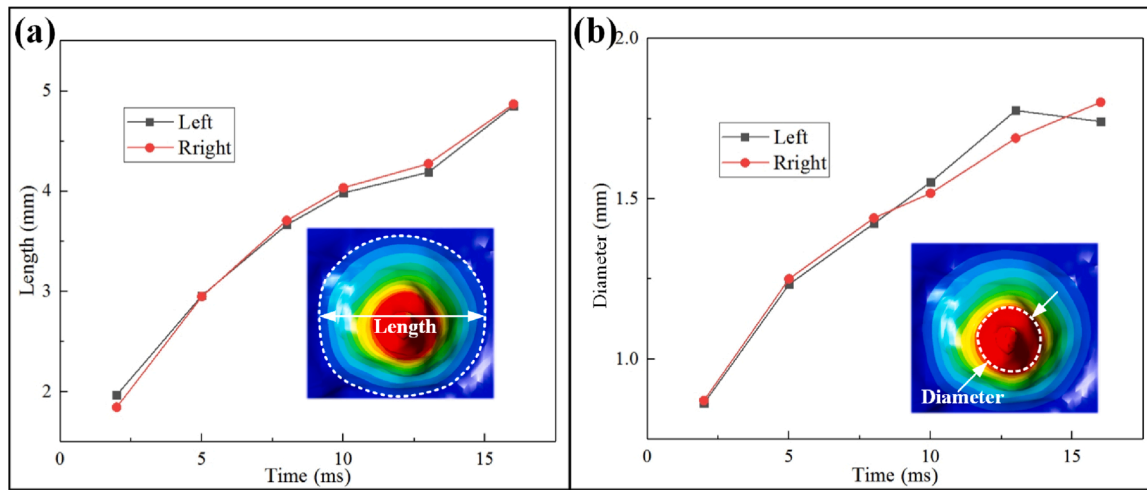


Fig. 10. The length of molten pool and opening diameter of keyhole at different times. (a) The length of the molten pool on both sides of stringer. (b) The opening diameter of keyhole on both sides of stringer.

After the coupled keyhole is formed, the morphology of molten pool surface and coupled keyhole wall at different moments are depicted in Fig. 11. As presented in Fig. 11(a–d), it is acquired that the part of the keyhole coupling area has the temperature slightly lower than boiling point of the base metal. Nonuniform temperature distribution might be caused by the energy attenuation of laser beam and the oscillation of the coupled keyhole in the primary coupling stage. During subsequent DLBSW process, not only the profile of coupled keyhole keeps oscillating, but its diameter gradually extends with the molten pool.

4.3. The melt flow behaviors

The melt flow behaviors at the molten pool surface and coupled

keyhole wall are presented in Fig. 12. Compared to the results in Fig. 12 (a–d), it is calculated that the distribution characteristics of the flow field at each moment are relatively similar. Liquid metal flows from the center to the edge of molten pool. The maximum flow velocity at each moment is between 1.269 m/s and 1.863 m/s, which is dozens of times of the welding speed.

Generally, the vapor recoil pressure at the bottom of the keyhole is the highest, which is the main reason for the highest fluid velocity near the position. However, it is found that the high flow velocity region is mainly distributed in the opening area of keyhole as well as the partial area of the keyhole wall. It is inferred that the result may be caused by the interaction of the recoil pressures on both sides when the two keyholes penetrate during the welding process. This interaction is

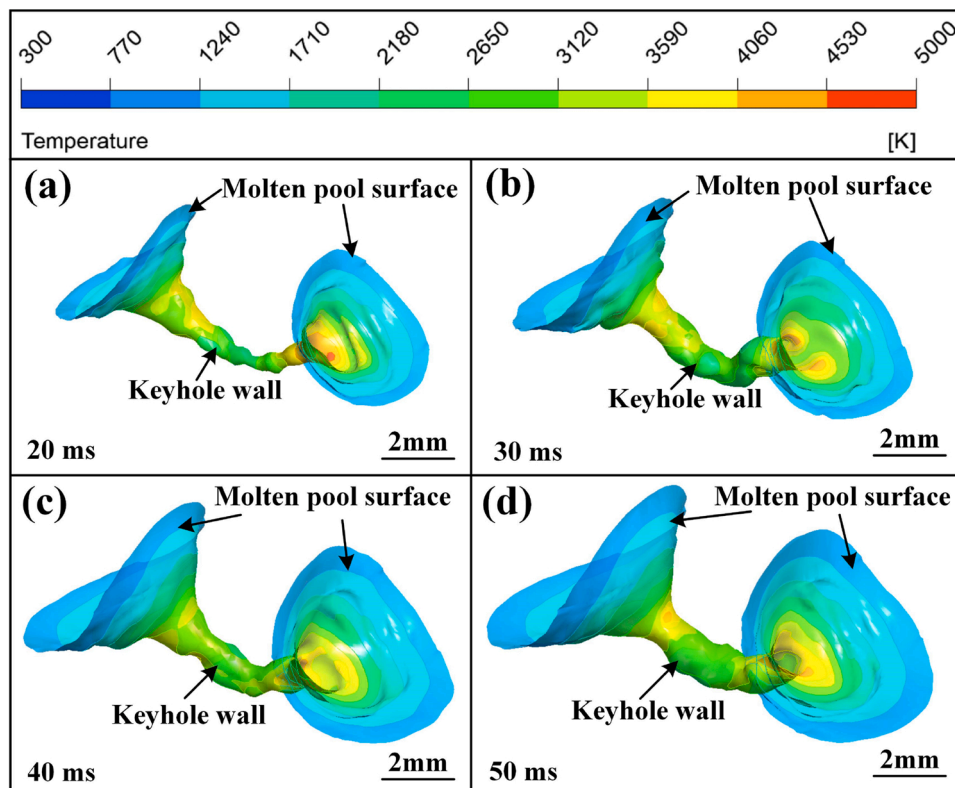


Fig. 11. Morphology of molten pool surface and coupled keyhole wall at different stages. (a) 20 ms. (b) 30 ms. (c) 40 ms. (d) 50 ms.

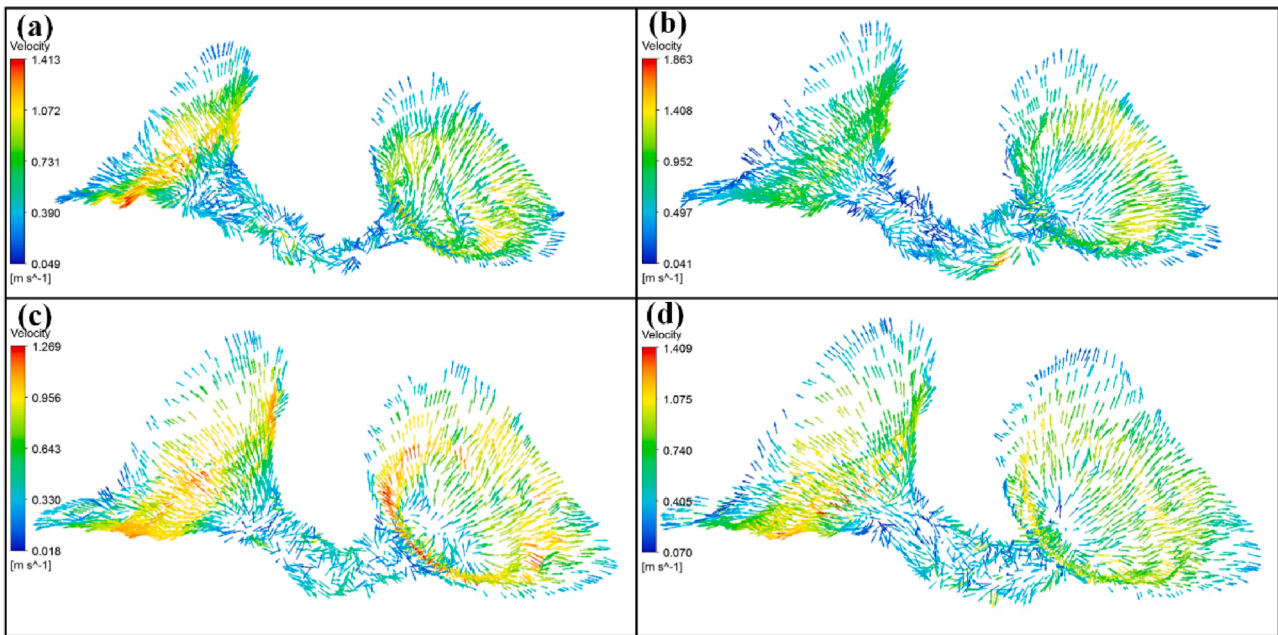


Fig. 12. The melt flow behaviors at molten pool surface and coupled keyhole wall at different stages. (a) 20 ms. (b) 30 ms. (c) 40 ms. (d) 50 ms.

significantly complicated so that some experiments and simulations are needed to implement in the subsequent studies. It is found that the velocity direction of melt flow on the keyhole wall is instable at different moments. Not only is there some liquid metal flowing in a direction parallel to the keyhole wall, but there is also some liquid metal flowing around the keyhole wall. It is inferred that the surface tension, the vapor recoil pressure in the keyhole and the keyhole oscillation interact together to cause the around metal flow and upward metal flow near the area of the keyhole wall during the process.

The morphology of molten pool on the left side during DLBSW process obtained by high-speed imaging is presented in Fig. 13. The profile of molten pool surface during DLBSW process is stable with few splashes generated. The position of keyhole in molten pool is relatively fixed, which is located in the bright laser area at the front of molten pool. It is observed that the keyhole moves with molten pool along the welding direction. The liquid metal accumulates towards the edge of molten pool due to the driving force of surface tension.

The surface flow behaviors with temperature distribution of the molten pool are shown in Fig. 14 when it moves to the middle of the model. It is observed from Fig. 14(b–c) that the morphology, melt flow behaviors and temperature distribution of the molten pool on both sides are significantly similar. The protrusion on the boundary of molten pool is formed, which is in good agreement with that in Fig. 13. Besides, the melt flows from center to the edge of molten pool. The extremely high temperature and dense isotherms are produced at keyhole area of molten pool. Isotherms at the rear of molten pool are significantly sparser compared to that at the front.

Aiming to analyze the characteristics of molten pool perpendicular to longitudinal section of molten pool surface, two planes shown in Fig. 15 (a) are selected for farther investigation. It is observed that the profiles of molten pool at the two sections are similar according to the results displayed in Fig. 15(b–c), respectively. The deepest location of molten pool slightly lags behind keyhole's back wall, which might be caused by the energy coupling of the two laser beams at the intersection of molten

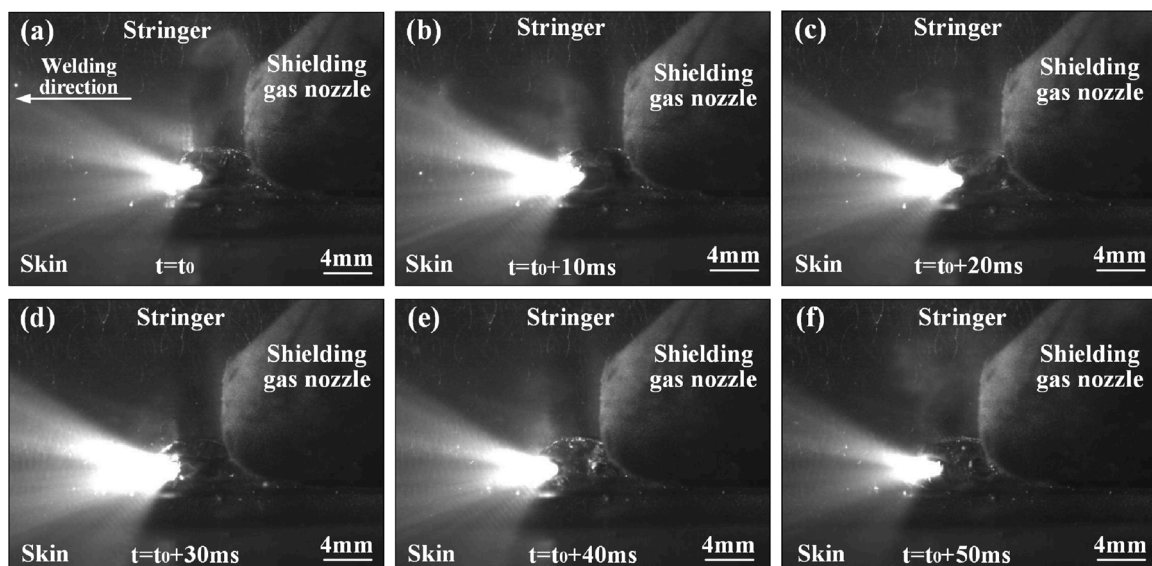


Fig. 13. The images of keyhole and molten pool dynamic behaviors obtained by high-speed camera.

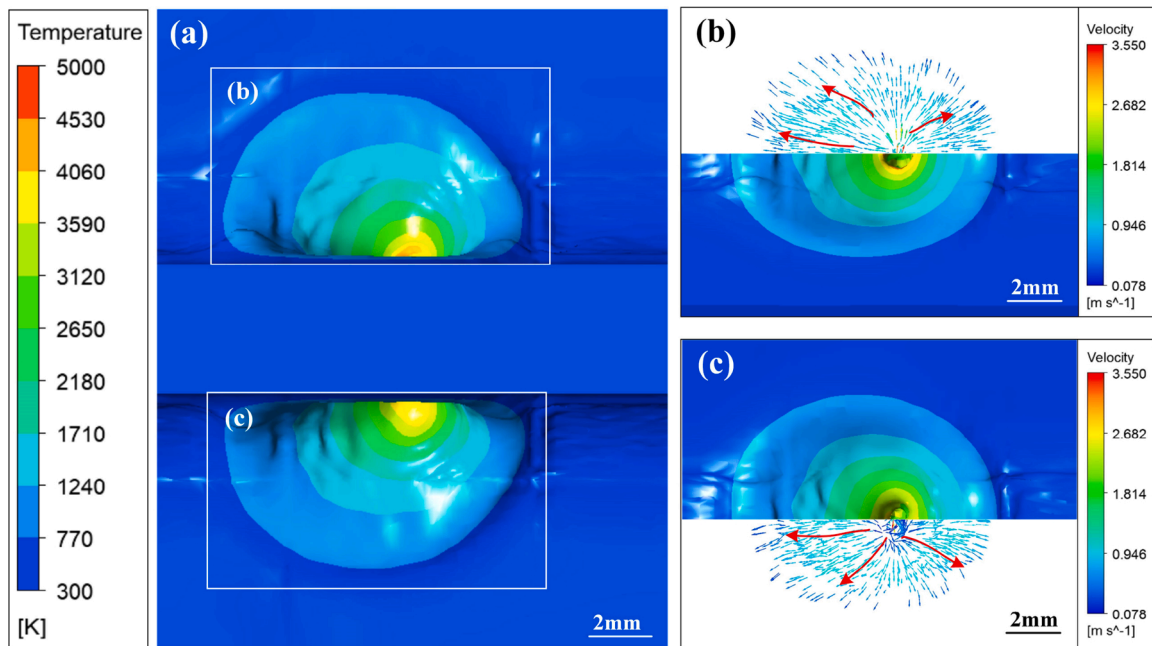


Fig. 14. The surface flow behaviors with temperature distribution of the molten pool on both sides of the T-joint.

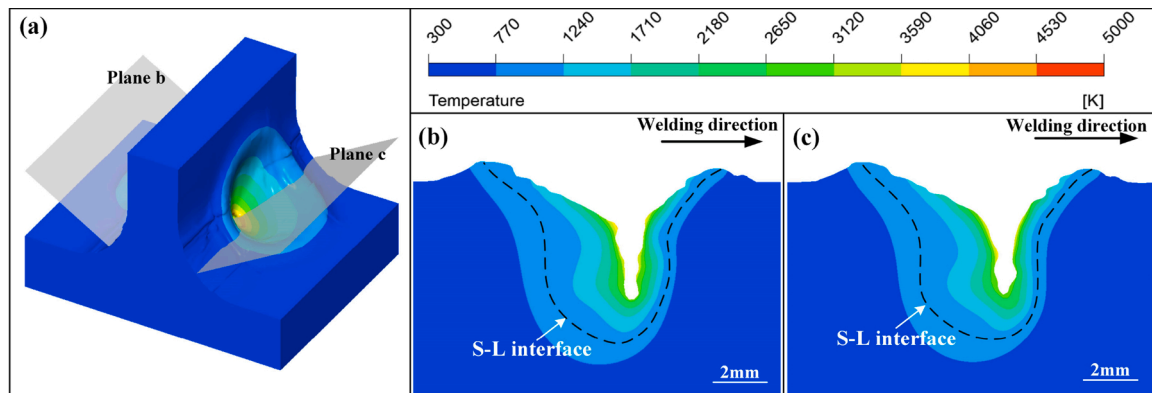


Fig. 15. The profile of molten pool and keyhole in longitudinal section. (a) Schematic diagram of longitudinal section position. (b) The profile of molten pool and keyhole on plane b marked in (a). (c) The profile of molten pool and keyhole on plane c marked in (a).

pool. The distance from the front wall of keyhole to the front boundary of molten pool is significantly shorter than the back wall of keyhole to the rear boundary of molten pool.

Moreover, the melt flow behaviors at the cross section and longitudinal section of molten pool selected from Fig. 16(a) are shown in Fig. 16 (b–d). It is obtained in Fig. 16(b) that the melt flow velocity on the keyhole wall and molten pool surface is significantly greater than that inside the molten pool. The vortices are produced near the upper and lower walls of the keyhole. It is formed by the combined effect of the vapor recoil pressure and thermal buoyancy, which drive the melt flows upward and merge with the liquid backflow. The distribution of the flow field of molten pool on plane c is almost similar to that on plane d. The flow directions of the liquid metal in different areas at rear region of molten pool are different, while they eventually converge at rear region near the keyhole wall and flow towards the tail of molten pool.

5. Conclusions

The keyhole coupling and melt flow behaviors of 2219 aluminum alloy T-joint during DLBSW process were investigated by CFD simulation in the present study. Based on the model verification, the coupling

process of keyhole and the evolution characteristics of coupled keyhole were explored. The characteristics of molten pool and flow behavior of liquid metal were studied. Some conclusions are summarized as follows:

- (1) Two keyholes and combined molten pools are rapidly generated at the initial stage of welding. The keyholes are gradually coupled to form a coupled keyhole around 16 ms after the beginning of the DLBSW process.
- (2) The profile of the coupled keyhole is continuously oscillated due to the effect of recoil pressure. The flow direction of liquid metal on the keyhole wall area mainly includes the flow parallel to keyhole wall and that around the keyhole wall.
- (3) With the progress of the welding process, the surface morphology and longitudinal section profile of molten pool on both sides of stringer are relatively symmetrical. The maximum depth of molten pool slightly lags behind the back wall of keyhole.
- (4) The flow velocities of the liquid metal on the keyhole wall and molten pool are significantly greater than that inside the molten pool. The vortices produced by vapor recoil pressure and thermal buoyancy are formed near the upper and lower regions of the keyhole wall.

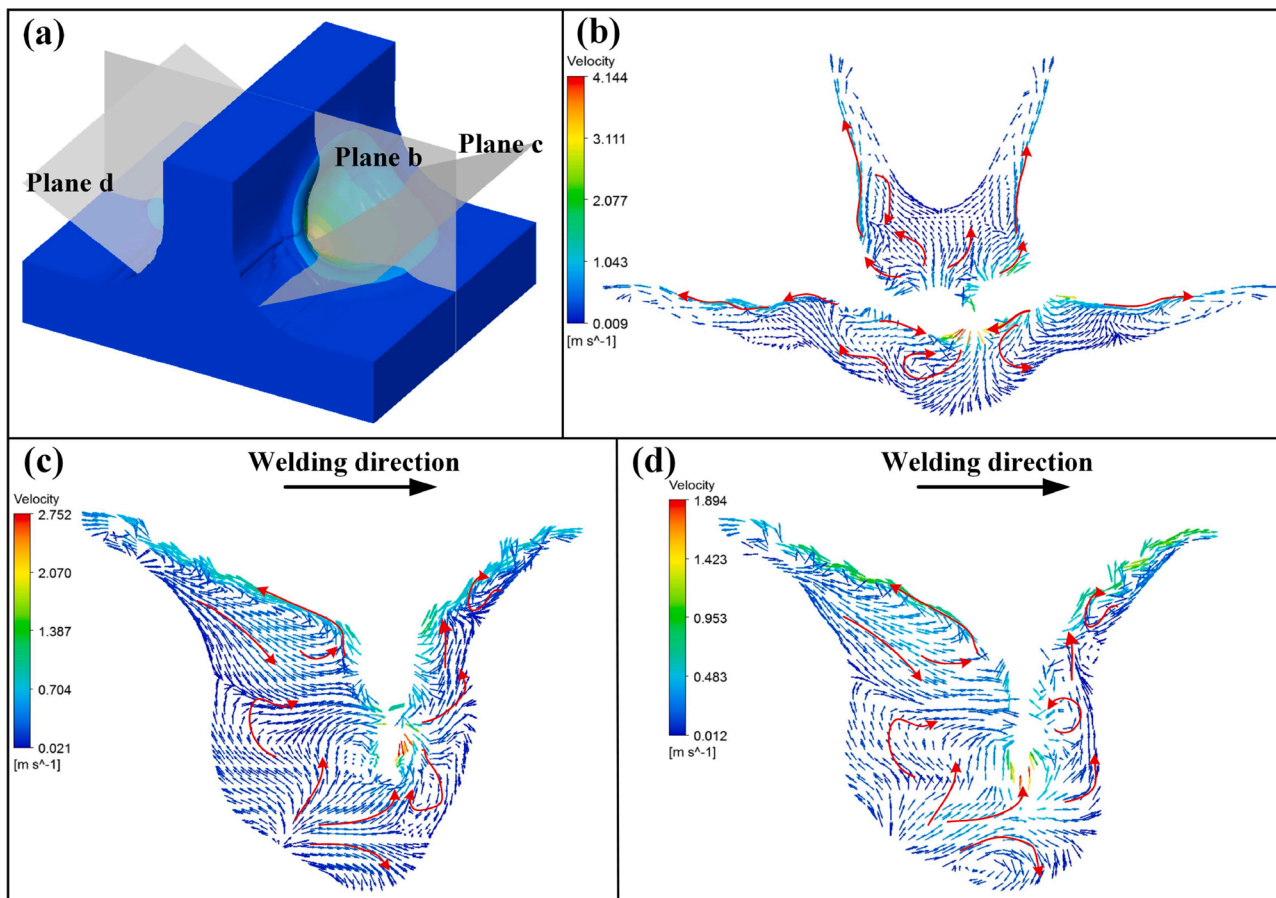


Fig. 16. The flow behaviors of liquid metal in molten pool. (a) Schematic diagram of section position. (b) The flow behaviors of liquid metal in molten pool on plane b marked in (a). (c) The flow behaviors of liquid metal in molten pool on plane c marked in (a). (d) The flow behaviors of liquid metal in molten pool on plane d marked in (a).

Declaration of Competing Interest

The authors report no declarations of interest.

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